Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
SEMESTER I
SESSION 2024/2025

COURSE NAME	:	OBJECT-ORIENTED PROGRAMMING
COURSE CODE	:	BIC 20904
PROGRAMME CODE	:	BIP / BIS / BIW / BIM
EXAMINATION DATE	:	FEBRUARY 2025
DURATION	:	3 HOURS
INSTRUCTIONS	:	1. ANSWER ALL QUESTIONS 2. THIS FINAL EXAMINATION IS CONDUCTED VIA ☐ Open book ☒ Closed book 3. STUDENTS ARE **PROHIBITED** TO CONSULT THEIR OWN MATERIAL OR ANY EXTERNAL RESOURCES DURING THE EXAMINATION CONDUCTED VIA CLOSED BOOK

TERBUKA

THIS QUESTION PAPER CONSISTS OF **FIVE (5)** PAGES

Q1 Read the statements carefully and state whether the statement is **TRUE** or **FALSE**.

(a) A method that returns nothing has no return type.

(1 mark)

(b) To compare two variables, both have to be in the same type.

(1 mark)

(c) Each element in a five-element array can hold five values.

(1 mark)

(d) To programmers, another word for object is instance.

(1 mark)

(e) When you increment a variable, you subtract 1 from it.

(1 mark)

(f) A class diagram consists of a rectangle divided into three sections, i.e., name, data and methods.

(1 mark)

(g) If a `>` `b` is false, then `a` `<=` `b` is always true.

(1 mark)

(h) The process of using the same method name to indicate different implementation is inheritance.

(1 mark)

(i) In the following pseudocode, `4 6 7` will be printed.

```
d = 4
e = 6
f = 7
while d > 7
      d = d + 1
      e = e - 1
endwhile
print d, e, f
```

(1 mark)

(j) The highest subscript you should use with a 10-element array is 10.

(1 mark)

TERBUKA

Q2 A Java program is to be developed for Astana Catering Company to keep records concerning the events it caters. The program comprises of an `Event` class that contains attributes `eventNumber`, `hostName`, and `evenDate`. It also contains the `guestNumber` that will be attending the event and the `meal` name that the event hosts will be serving. All information should be kept as `private`.

A no argument constructor is to be included in the class. Additionally, the class comes with mutator and accessor methods for each of the information, as well as the other two methods to calculate total price and total price after discount; with fixed price of meal per guest, i.e. RM15.00 for any type of meal. For the total price calculation, method headers are given as follows:

```
public double calcTotal(), and
public double calcTotalDiscount()
```

(a) Identify any **FOUR (4)** member variables of the class.

(2 marks)

(b) Declare the final variable.

(2 marks)

(c) Illustrate the class representation using UML notation. (Note: You do not have to include the final variable in the class representation)

(5 marks)

(d) Based on your answer in **Q2(c)**,

(i) Write the method to calculate total price after discount. Discount from the total price is given based on **Table Q2.1**.

Table Q2.1 Given discount in percentage (%) based on number of guests

Number of guests	Discount from the total price
0 - 99	-
100 – 199	5%
200 - 499	10%
≥500	20%

(5 marks)

(ii) Write the accessor method for `guestNumber`.

(2 marks)

TERBUKA

(iii) Another class called `Main` is used to test the class and will contain a `main()` method. Inside the method, create an instance of `Event` class with the following initiation to the instance:

```
1, "Mr. Ali bin Abu", "1 Oct 2026", "Nasi Beriani", 500
```

And then, display the event information again. Sample of the output is as follows:

```
Event number: 1
Event host: Mr. Ali bin Abu
Event date: 1 Oct 2026
Menu: Nasi Beriani
Number of guests: 500
TOTAL PRICE (RM): 7500.0
TOTAL PRICE AFTER DISOCUNT (RM): 6000.0
```

(4 marks)

Q3 Answer **Q3(a) – Q3(b)** based on **Figure Q3.1** and **Table Q3.1**.

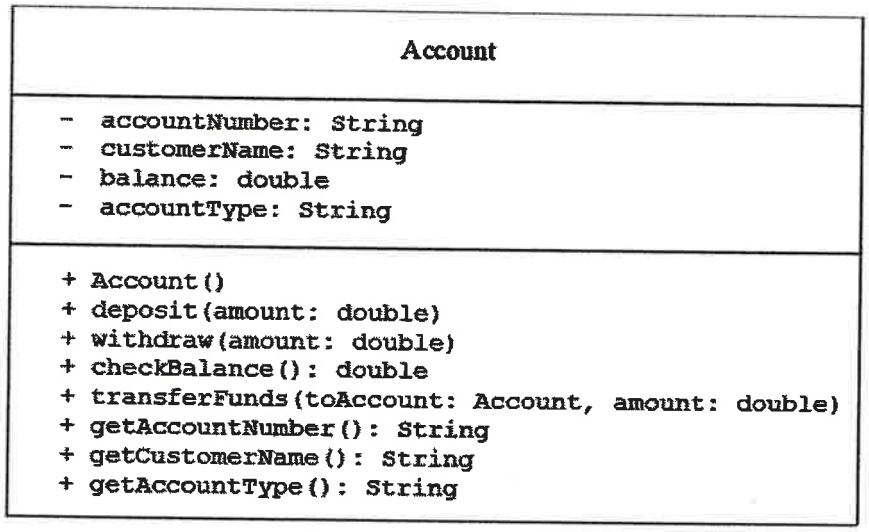


Figure Q3.1 `Account` class

(a) Write a complete program based on the UML class diagram given in **Figure Q3.1**.

(16 marks)

(b) Write a program named `BankAccount` that create and displays **TWO (2)** objects of `Account` with the value of the attributes as shown in **Table Q3.1**.

Table Q3.1 Value for `Account` objects' attribute

accountNumber	customerName	balance	accountType
251239	Samantha	RM 10,00.00	"Saving"
768856	Junaida	RM 5,000.00	"Current"

TERBUKA

(4 marks)

Q4 Answer all the following questions.

(a) Draw a class hierarchy using inheritance to organize the following classes: `Patient`, `outpatient` and `inpatient`.

(3 marks)

(b) Using Java, implement all classes according to the class hierarchy with the following details:

(i) Each outpatient shall have the following attributes: name, patient id number, consultation fee, diagnosis, prescription, last appointment date, and next appointment date. Each outpatient has one method `ShowPatient` without parameter to display output of the outpatient.

(8 marks)

(ii) Each inpatient shall have the following attributes: name, patient id number, consultation fee, diagnosis, prescription, warded date, ward number, warded fee and specialist name. Each inpatient has one method `ShowPatient` without parameter to display output of the inpatient.

(8 marks)

(iii) Implement the constructor for the Patient class with common parameters and one method `ShowPatient` without parameter that display the general information details of the patient.

(6 marks)

(c) Implement the `main` method to test the patient classes.

(5 marks)

- END OF QUESTIONS -

TERBUKA